Course Project

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2024-12-07

Specifications

The purpose of the data I am using the Titanic data set downloaded from this kaggle.

1. The response variable: `Survived` (0 if dead, 1 if survived)
2. The protection purpose of the data: I want to predict if a person on the Titanic would survive or die during the disaster
3. The protection variables:
   - `Pclass`: passenger class (1, 2, 3)
   - `Sex`: female, 0 male
   - `Age`: (age in years)
   - `Fare`: (fare paid in shillings/pounds sterlin)
4. The evaluation metric: I will use the determine accuracy. R^2

```
# set path for rounding testing
options(digits = 4)

# load data
TitanicTrain = read.csv("train.csv")
TitanicTestData = read.csv

## [1] 418

TitanicTrain = read.csv("train.csv")
TitanicTestData = read.csv

## [1] 418

# check numeric variables and factored var summaries
TitanicTrainNumeric = TitanicTrain[sapply(TitanicTrain, is.numeric)]
summaryTitanicTrainNumeric = sapply(TitanicTrainNumeric, function(x) c(mean=mean(x,na.rm=TRUE), sd=sd(x,na.rm=TRUE), min=min(x,na.rm=TRUE), q1=quantile(x,0.25,na.rm=TRUE), median=median(x,na.rm=TRUE), q3=quantile(x,0.75,na.rm=TRUE), max=max(x,na.rm=TRUE)))

TitanicTrainNumeric = TitanicTrain[sapply(TitanicTrain,"Pclass", "Sex", "Age", "SibSp", "Parch", "Fare")]
summaryTitanicTrainNumeric = sapply(TitanicTrainNumeric, function(x) c(mean=mean(x,na.rm=TRUE), sd=sd(x,na.rm=TRUE), min=min(x,na.rm=TRUE), q1=quantile(x,0.25,na.rm=TRUE), median=median(x,na.rm=TRUE), q3=quantile(x,0.75,na.rm=TRUE), max=max(x,na.rm=TRUE)))

# check missing var values with the dplyr
TitanicMissingCheck = sapply(TitanicTrain, function(x) c(NA=sum(is.na(x)), Percent=mean(is.na(x))*100))
TitanicMissingCheck

# check for missing var values in test data
TitanicMissingCheckTest = sapply(TitanicTestData, function(x) c(NA=sum(is.na(x)), Percent=mean(is.na(x))*100))
TitanicMissingCheckTest

# factor survival, sex, embarked, pclass, sibsp and parch variables
TitanicTrain$Survived = factor(TitanicTrain$Survived, levels=c(0,1), labels=c("Dead", "Survived"))
TitanicTrain$Sex = factor(TitanicTrain$Sex, levels=c("female", "male"))
TitanicTrain$Embarked = factor(TitanicTrain$Embarked, levels=c("C", "Q", "S"), labels=c("Cherbourg", "Queenstown", "Southampton"))
TitanicTrain$Pclass = factor(TitanicTrain$Pclass, levels=c(1,2,3), labels=c("First", "Second", "Third"))
TitanicTrain$SibSp = factor(TitanicTrain$SibSp)
TitanicTrain$Parch = factor(TitanicTrain$Parch)

# factor test data
TitanicTestData$Sex = factor(TitanicTestData$Sex, levels=c("female", "male"))
TitanicTestData$Pclass = factor(TitanicTestData$Pclass, levels=c(1,2,3), labels=c("First", "Second", "Third"))
TitanicTestData$SibSp = factor(TitanicTestData$SibSp)
TitanicTestData$Parch = factor(TitanicTestData$Parch)

# create new data set with factored variables
TitanicTrainFactored = TitanicTrain
TitanicTestDataFactored = TitanicTestData

# check factored var summaries
summaryTitanicTrainFactored = sapply(TitanicTrainFactored, function(x) c(NA=sum(is.na(x)), Percent=mean(is.na(x))*100))
summaryTitanicTrainFactored

# check factored var summaries
summaryTitanicTestDataFactored = sapply(TitanicTestDataFactored, function(x) c(NA=sum(is.na(x)), Percent=mean(is.na(x))*100))
summaryTitanicTestDataFactored

# create new data set without factored variables
TitanicTrainNoFactor = TitanicTrain[sapply(TitanicTrain, function(x) !is.factor(x))]
TitanicTrainNoFactor

# create new data set without factored variables
TitanicTestDataNoFactor = TitanicTestData[sapply(TitanicTestData, function(x) !is.factor(x))]
TitanicTestDataNoFactor

# create new data set with numeric variables
TitanicTrainNumeric = TitanicTrain[sapply(TitanicTrain, function(x) is.numeric(x))]
TitanicTrainNumeric

# create new data set with numeric variables
TitanicTestDataNumeric = TitanicTestData[sapply(TitanicTestData, function(x) is.numeric(x))]
TitanicTestDataNumeric

# create new data set with factored variables
TitanicTrainFactor = TitanicTrain[sapply(TitanicTrain, function(x) is.factor(x))]
TitanicTrainFactor

# create new data set with factored variables
TitanicTestDataFactor = TitanicTestData[sapply(TitanicTestData, function(x) is.factor(x))]
TitanicTestDataFactor
```